a
Analysis pipeline
1
ITC Database
100 studies
11,852 people
1.17 M choices
2
Cleaning
database exclusions
SS > 0, LL > SS, LL later
≥ 10 choices per person
3
Study selection
41 qualified
40 fitted
39 pooled
4
Hierarchical Bayesian fitting
PyMC + NumPyro
NUTS sampling
three models per study
5
Evaluation
PSIS-LOO model comparison
meta-analytic pooling
model recovery
b
Three models of how reward magnitude r shapes choice
Same data and choice rule, compared by out-of-sample fit
H2 Descriptive
Discount rate and choice noise both vary freely with reward magnitude.
$\log k = c_k - m_k \log r$
$\log \alpha = c_\alpha + m_\alpha \log r_\Sigma$
P(choose later)
relative value difference
reward
High
Medium
Low
fitted: precision $\propto r^{0.31}$
R2 Rational inattention
Noise is set by an attention cost, so the model fixes how noise scales.
$k = 1/(\beta r)$, α fixed by β
relative value difference
reward
High
Medium
Low
predicted: precision $\propto r^{1.5}$
R2F R2 + free noise slope
R2 with the noise slope g_1 left free ($g_1 = 0$ recovers R2).
$\alpha = \alpha_{R2} \exp(g_0 + g_1 \log r_\Sigma)$
relative value difference
reward
High
Medium
Low
fitted: precision $\propto r^{0.34}$
c
Out-of-sample fit
ΔELPD vs R2, 37 studies
H2
+21,378
R2F
+13,233
R2
0 (reference)
0
10k
20k
30k
ΔELPD vs R2
H2 beats R2 in 39 of 39 studies;
R2F beats R2 in 37 of 37.